

High-Frequency Market Making via Reinforcement Learning under Different Volatility Regimes

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Contents

Abstract.....	5
Symbols, Abbreviations, and Glossary.....	5
1 Introduction	7
1.1 Problem Statement and Motivation	7
1.2 Research Objective	7
1.3 Background and Related Work	7
1.4 Contributions	8
1.5 Scope and Limitations	9
1.6 Report Structure	9
2 Mathematical Formulation	9
2.1 Synthetic Mid-Price Process.....	9
2.2 Markov Volatility-Regime Process	9
2.3 Realized Volatility Signal and Regime Detection	9
2.3.1 Illustrative Synthetic Environment Path	10
2.4 Limit-Order Fill Model.....	10
2.5 Quote Parameterization	11
2.6 Inventory, Cash, Equity, and Reward	11
2.7 Avellaneda-Stoikov Baseline	12
2.8 PPO Objective	12
3 Use-Case Scenario.....	12
3.1 Synthetic HFMM Scenario.....	12
3.2 Trading Agents and Strategy Variants.....	13
3.3 Information-Design Question	13
4 System Architecture.....	13
4.1 Run Lifecycle and Reproducibility Layer	13
4.2 Synthetic Market and Detector Layer	13
4.3 Gymnasium Environment Layer.....	13
4.4 Strategy and Training Layer	14
4.5 Evaluation and Evidence Layer	14
4.6 Methodology Pipeline Illustration	14
5 Layered Experimental Model	14

5.1 Layer 1: Market State Generation	14
5.2 Layer 2: Signal Construction.....	14
5.3 Layer 3: Observation Encoding	15
5.4 Layer 4: Action and Execution.....	15
5.5 Layer 5: Learning and Evaluation	15
6 Algorithm Specification and Pseudocode	15
6.1 Plain-Language Overview	15
6.2 Synthetic Market and Regime Generation	15
6.3 Market-Making Environment Step	15
6.4 PPO Training and OOS Evaluation Protocol	15
6.5 Experiment 5: Signal-Informativeness Sweep	16
6.6 Reference Implementation Map.....	16
6.7 Complexity Analysis	16
6.8 Executable Example Commands	17
Commands	17
7 Experimental Setup.....	17
7.1 Canonical Configuration Protocol	17
7.2 Data Generation and Preprocessing	17
7.3 Train/Test Split.....	17
7.4 Strategy Variants.....	17
7.5 Detector Variants	18
7.6 PPO Hyperparameters	18
7.6.1 Hyperparameter Selection Protocol	18
7.7 Protected Evidence and No-Rerun Policy	19
8 Results and Visualisation.....	19
8.1 Experiment 1: Main Out-of-Sample Evaluation	19
8.2 Five-Variant Ablation	20
8.3 Detector Robustness.....	22
8.4 Reward-Shaping and Misspecification Checks.....	23
8.5 Experiment 5: Signal-Informativeness Sweep	24
9 Evaluation Metrics	27
9.1 Final Equity.....	27
9.2 Sharpe-Like Ratio	27

9.3 Inventory Tail Risk	28
9.4 Fill Rate.....	28
9.5 Paired t-Tests	28
9.6 TOST Equivalence Tests.....	28
10 Discussion.....	28
10.1 Main Interpretation: Signal Redundancy	28
10.2 Why the Result Is Not a Weak Null	29
10.3 Synthetic-Market Boundary	29
10.4 Risks to Interpretation	29
10.5 Future Work	30
11 Reproducibility Checklist.....	30
12 Conclusion.....	32
References	32
Appendix A Code Appendix.....	33
Canonical commands	33
Appendix B Unit Tests and Validation Checks	33
Appendix C Extended Evidence Tables	34
Appendix D Author Contribution Statement	37

Abstract

This report studies whether explicit volatility-regime information improves a Proximal Policy Optimization (PPO)-based high-frequency market-making (HFMM) agent in a controlled synthetic limit-order-book (LOB) environment. The simulator uses an arithmetic Brownian-motion mid-price, a sticky Markov volatility-regime process, rolling realized-volatility detection, and Poisson-arrival fills. The PPO agent chooses bid and ask quotes through a discrete half-spread and skew parameterization, and it is compared with a fixed-spread strategy and an Avellaneda-Stoikov analytical baseline.

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy. The evidence comes from the canonical frozen experiments: out-of-sample evaluation, detector robustness, oracle-label ablations, regime-conditional reward shaping, mild model misspecification, and a signal-informativeness sweep. In the five-variant ablation, σ_{only} achieved the highest mean Sharpe-like value (0.753), while σ_{full} did not provide a statistically significant or practically meaningful improvement over $\sigma_{\hat{}}$ alone (paired t-test $p = 0.115$; TOST $+/-0.10$ $p = 0.00067$; 90% CI $[-0.001, +0.063]$). The strongest interpretation is therefore conditional and synthetic-market bounded: the explicit regime label mostly repackages volatility information already present in the continuous signal. The report makes no production-deployment claim and does not claim to prove the internal PPO mechanism.

Symbols, Abbreviations, and Glossary

Symbol	Meaning
S_t	Mid-price at time step t .
q_t	Inventory after step t .
X_t	Cash account after step t .
W_t	Marked-to-market wealth or equity, $W_t = X_t + q_t S_t$.
$\sigma_{\hat{}}$	Rolling realized-volatility estimate observed by the PPO policy.
R_t	Reward at time t .
h	Quoted half-spread in ticks.
m	Quote skew in ticks.
A, k	Poisson fill-intensity scale and decay parameters.
η	Inventory penalty coefficient.

γ	Avellaneda-Stoikov inventory-risk aversion parameter.
τ	Remaining horizon.
$\lambda(\delta)$	Poisson fill intensity at quote distance δ .
$\delta, \delta_{\text{bid}}, \delta_{\text{ask}}$	Quote distance variables.
$\rho_t(\theta)$	PPO probability ratio.
A_{hat}_t	Advantage estimate.
ϵ_t	Standard-normal innovation in the mid-price process.
z_t	Latent volatility regime.
P_{ij}	Markov transition probability.

Abbreviation	Meaning
ABM	Arithmetic Brownian motion.
AS	Avellaneda-Stoikov baseline.
HFMM	High-frequency market making.
OOS	Out-of-sample evaluation.
PPO	Proximal Policy Optimization.
RV	Realized volatility.
TOST	Two One-Sided Tests equivalence procedure.
WP	Work package.

Term	Definition
Regime-aware PPO	PPO policy variant that receives a categorical regime one-hot channel.
Regime-blind PPO	PPO policy variant that omits the regime one-hot but still observes σ_{hat} .
σ_{only}	Ablation variant using the continuous volatility signal without categorical regime labels.

combined	Ablation variant using $\sigma_{\hat{}}$ and estimated categorical regime labels.
oracle_full	Ablation variant using $\sigma_{\hat{}}$ and the true regime label.

1 Introduction

1.1 Problem Statement and Motivation

Market makers continuously choose bid and ask quotes while balancing spread capture against inventory risk. When volatility changes, the same quote width can become either too passive or too exposed. A natural response is to provide a learning agent with regime labels such as low, medium, and high volatility. The central question is whether that categorical label adds decision-relevant information when the policy already observes a continuous volatility estimate.

1.2 Research Objective

The objective is to test the incremental value of explicit categorical volatility-regime labels in PPO market making under a controlled synthetic environment. The comparison is intentionally information-design focused: the regime label is evaluated against a continuous $\sigma_{\hat{}}$ signal, not against an observation space with no volatility information.

1.3 Background and Related Work

The literature relevant to this thesis falls into four connected strands: classical inventory-aware market making, reinforcement-learning market making, volatility regimes and non-stationarity, and statistical interpretation of small or practically negligible differences. The purpose of the review is not to claim a new analytical quoting rule or a new reinforcement-learning algorithm. It is to locate a narrower information-design question inside an established market-making literature.

Classical market-making models provide the inventory-risk and quote-placement foundation. A market maker earns spread income by posting bid and ask quotes, but stochastic execution creates inventory exposure whose value changes with the mid-price. Avellaneda and Stoikov (2008) formalize this tradeoff through a reservation price and spread term that depend on inventory, risk aversion, volatility, remaining horizon, and order-arrival intensity. Gueant, Lehalle and Fernandez-Tapia (2013) extend this inventory-control view and provide tractable approximations for optimal quotes under inventory constraints.

This analytical literature motivates the thesis design in two direct ways: it explains why inventory, volatility, quote distance, and fill intensity are central state variables, and it supplies the Avellaneda-Stoikov baseline used as a non-learning comparator. Related preprints by Fodra and Labadie (2012, 2013) are useful additional context for inventory-constrained and state-dependent market-making models, but they are treated cautiously as preprints. The thesis does not attempt to derive a new closed-form quoting rule; it uses the analytical tradition to frame the simulation and benchmark design.

A second strand replaces closed-form rules with learned policies. Spooner et al. (2018) demonstrate that reinforcement learning can learn market-making behavior in a simulated limit-order-book setting and show why reward design and inventory control matter. Spooner and Savani (2020) extend the discussion toward robustness under adversarially varied market conditions. These studies support the use of simulator-based evaluation for learned quoting policies, while also showing that market-making performance can be sensitive to the assumptions used to generate training and evaluation episodes.

Deep reinforcement-learning studies with market signals are closest to the present information-design question. Gasperov and Kostanjcar (2021) study market making with signals through deep reinforcement learning; Gasperov et al. (2021) review reinforcement-learning approaches to optimal market making; and Gasperov and Kostanjcar (2022) train agents in a richer Hawkes-process limit-order-book simulator. This literature motivates learned quoting, state representation, reward shaping, robustness checks, and simulator-based evidence. The present thesis is narrower: PPO is used as a controlled testbed for whether one additional volatility-regime channel has incremental value, not as a claim of algorithmic optimality.

Volatility regimes and non-stationarity are economically meaningful. A high-volatility state can increase inventory risk, change fill-risk tradeoffs, and make a quote width that was appropriate in a calm period too aggressive in a turbulent period. The thesis therefore models a latent three-state Markov volatility process and a causal rolling realized-volatility detector. This design should not be read as a claim that volatility regimes are irrelevant. The tested question is more specific: whether categorical regime labels add incremental value once the policy already observes a continuous realized-volatility estimate.

The key state-representation issue is redundancy. The continuous signal $\hat{\sigma}$ gives the PPO policy a graded estimate of recent realized volatility. A low/medium/high label derived from related volatility information may help if it summarizes latent structure that $\hat{\sigma}$ does not expose, but it may add little if the continuous signal already contains the useful variation for quote placement. This gives the exact gap addressed by the thesis: When a PPO market-making policy already observes a continuous realized-volatility estimate, does adding an explicit categorical volatility-regime label improve out-of-sample performance in a controlled synthetic HFMM environment?

Finally, the empirical interpretation requires care because the thesis evaluates an incremental signal, not the existence of volatility effects in general. Lakens (2017) and Lakens, Scheel and Isager (2018) motivate equivalence testing as a way to distinguish a merely non-significant difference from evidence that a difference is small relative to a pre-specified practical bound. The TOST framing is therefore used as background for the later statistics section; it does not by itself prove that labels contain no information.

1.4 Contributions

1. A controlled HFMM simulator with Poisson-arrival fills, fees, latency, and Markov volatility regimes.
2. A Gymnasium environment in which PPO controls quote half-spread and skew.
3. An OOS comparison of fixed-spread, Avellaneda-Stoikov, regime-aware PPO, and regime-blind PPO strategies.
4. A five-variant ablation separating continuous volatility information from estimated and oracle regime labels.

5. Detector, reward-shaping, misspecification, and signal-informativeness checks that bound the interpretation.

1.5 Scope and Limitations

The thesis is a controlled synthetic-market study. It does not claim production deployment viability, real-order-book external validity, or a proven PPO representation mechanism. The result is conditional on the simulator, signal design, PPO hyperparameters, and degradation calibration tested here.

1.6 Report Structure

Sections 2-6 define the mathematical model, use case, architecture, layered model, and algorithms. Sections 7-9 describe the experimental setup, results, and metrics. Sections 10-12 discuss the interpretation, reproducibility, and conclusion. Appendices provide code maps, sanity checks, and extended evidence.

2 Mathematical Formulation

2.1 Synthetic Mid-Price Process

$$(1) S_{t+1} = S_t + \sigma_t \sqrt{dt} \epsilon_t, \quad \epsilon_t \sim N(0, 1)$$

Here S_t is the mid-price, σ_t is the step volatility in ticks or price units according to the simulator configuration, dt is the time step, and ϵ_t is an independent standard-normal shock.

2.2 Markov Volatility-Regime Process

$$(2) P(z_{t+1}=j | z_t=i) = P_{ij}, \quad z_t \in \{L, M, H\}$$

Here z_t is the latent volatility regime at step t , L/M/H denote low, medium, and high volatility, and P_{ij} is the sticky transition probability from regime i to regime j .

$$(2a) P = \begin{bmatrix} 0.9967 & 0.0023 & 0.0010 \\ 0.0042 & 0.9917 & 0.0041 \\ 0.0010 & 0.0030 & 0.9960 \end{bmatrix}$$

Rows correspond to the current regime L/M/H, and columns correspond to the next regime L/M/H. The large diagonal probabilities make the regimes persistent, or sticky, so volatility states tend to last for many steps rather than switching erratically.

This persistence matters because rolling realized-volatility detection is only meaningful when regime-dependent volatility differences last long enough to be observed through a finite window.

The regime controls the volatility multiplier applied to the base σ parameter. The canonical full experiments use σ multipliers [0.6, 1.0, 1.8] for L, M, and H.

2.3 Realized Volatility Signal and Regime Detection

$$(3) \hat{\sigma}_t = \sqrt{\left(\frac{1}{w}\right) \sum_{i=t-w+1}^t (\Delta S_i)^2}$$

Here $\hat{\sigma}_t$ is the rolling realized-volatility signal, w is the rolling window length, and ΔS_i is the mid-price increment over step i .

Estimated regime labels are assigned by thresholding $\sigma_{\hat{}}$ after a warmup period. The main reported training and evaluation pipelines use the causal $rv_baseline$ detector. The rv_dwell detector is retained only as an auxiliary/offline robustness comparison, while the HMM detector is an additional robustness variant.

2.3.1 Illustrative Synthetic Environment Path

This subsection visualizes one deterministic synthetic path generated under the canonical HFMM environment configuration. The figure is included to make the simulated market environment transparent: the mid-price evolves under a regime-switching arithmetic Brownian-motion process, the realized-volatility signal is computed from rolling mid-price changes, and the latent Markov regime switches between low, medium, and high volatility states. The dashed vertical line marks the chronological 70/30 train-test split used in the out-of-sample protocol. The figure is illustrative of the methodology only and is not a new performance result.

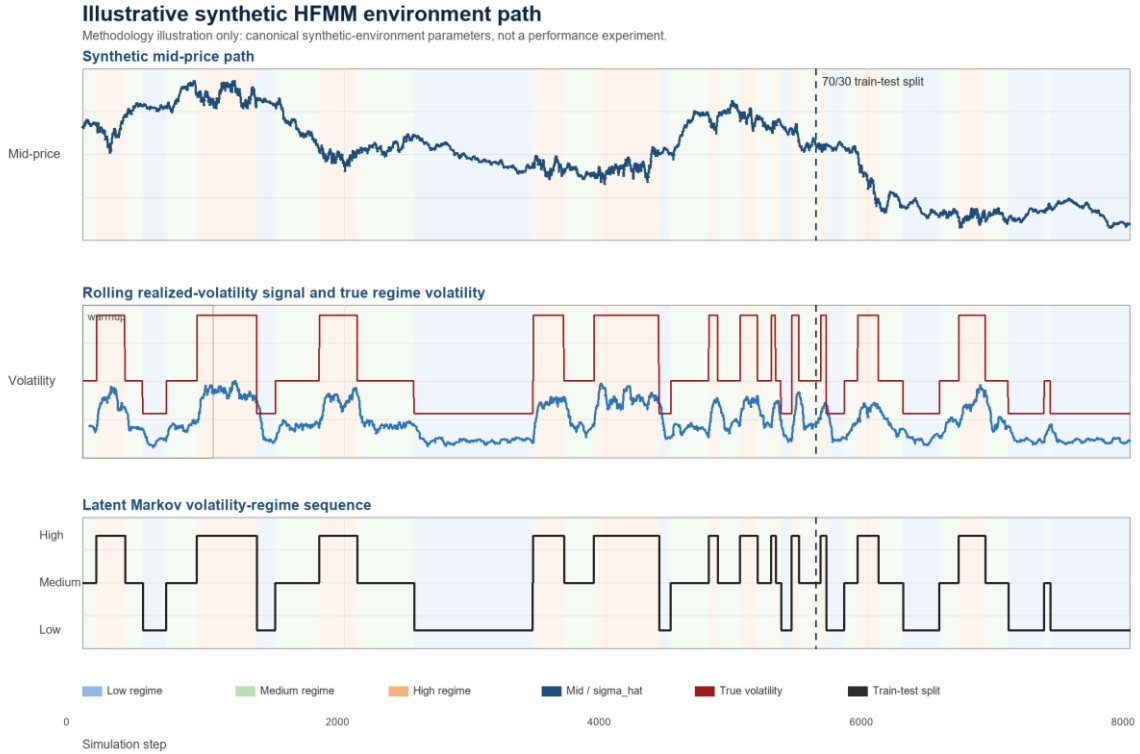


Figure 2.1. Illustrative synthetic HFMM environment path. The upper panel shows the generated mid-price path. The middle panel compares the rolling realized-volatility signal observed by the policy with the true regime-dependent volatility level. The lower panel shows the latent Markov volatility-regime sequence. The dashed vertical line marks the chronological 70/30 train-test split used for out-of-sample evaluation. This figure is generated by `scripts/figures/gen_synthetic_environment_figure.py` and is a methodology illustration, not a regenerated performance artifact.

2.4 Limit-Order Fill Model

$$(4) \lambda(\delta) = A \exp(-k \delta)$$

Here $\lambda(\delta)$ is the fill intensity at quote distance δ , A is the baseline arrival scale, and k controls the exponential decay as quotes move farther from the mid-price.

Intuitively, quotes closer to the mid-price fill more often because they are more attractive to incoming market orders. Quotes farther away earn more spread if they fill, but the exponential intensity makes those fills less likely.

$$(5) P(\text{fill} \mid \delta) = 1 - \exp(-\lambda(\delta) dt)$$

Here $P(\text{fill} \mid \delta)$ is the per-step fill probability and dt is the simulator step length.

2.5 Quote Parameterization

$$(6) h = h_{\text{idx}} + 1, \quad m = m_{\text{idx}} - 2$$

Here h_{idx} and m_{idx} are the two discrete PPO action components, h is the half-spread in ticks, and m is the quote skew in ticks.

The half-spread h controls quote width and aggressiveness: small h is more aggressive and large h is more passive. The skew m shifts the bid and ask distances asymmetrically, allowing the policy to lean against inventory or other state pressure while still quoting both sides.

$$(7) \delta_{\text{bid}} = \max(1, h + m), \quad \delta_{\text{ask}} = \max(1, h - m)$$

Here δ_{bid} and δ_{ask} are the bid and ask quote distances in ticks. The max operator enforces a minimum quote distance of one tick.

2.6 Inventory, Cash, Equity, and Reward

$$(8) q_{t+1} = q_t + F_t^{\text{bid}} - F_t^{\text{ask}}$$

Here q_t is inventory, F_t^{bid} is the bid-side fill indicator or count, and F_t^{ask} is the ask-side fill indicator or count.

$$(9) X_{t+1} = X_t - F_t^{\text{bid}} P_t^{\text{bid}} + F_t^{\text{ask}} P_t^{\text{ask}} - \text{fees}_t$$

Here X_t is cash, P_t^{bid} and P_t^{ask} are the executed bid and ask prices, and fees_t denotes transaction costs.

$$(10) W_t = X_t + q_t S_t$$

Here W_t is mark-to-market wealth or equity, X_t is cash, q_t is inventory, and S_t is the mid-price.

The equity measure is marked to the mid-price for consistency across simulated strategies. This is an evaluation convention inside the controlled simulator and should not be interpreted as immediately liquidatable wealth in a real limit-order book. In practice, liquidating inventory would generally require crossing the spread, facing queue priority and available depth, and potentially incurring additional adverse-selection or market-impact costs. These effects are part of the real-market boundary discussed later, not part of the current controlled experiment.

$$(11) R_t = W_{t+1} - W_t - \eta q_{t+1}^2$$

Here R_t is the reward and η is the inventory-penalty coefficient. Fees are included in the cash update and are not counted a second time in the reward.

The ηq^2 term discourages the agent from earning apparent PnL by accumulating large directional inventory. It connects the learning objective to market-making risk control rather than pure speculation on the mid-price path.

2.7 Avellaneda-Stoikov Baseline

$$(12) \quad r_t = S_t - q_t \gamma \sigma^2 \tau$$

Here r_t is the reservation price, γ is inventory-risk aversion, σ is volatility, and τ is remaining horizon.

$$(13) \quad \Delta_{AS} = 0.5 \gamma \sigma^2 \tau + (1 / \gamma) \log(1 + \gamma / k)$$

Here Δ_{AS} is the AS half-spread and k is the same fill-intensity decay parameter used in the execution model. The implemented deltas are clipped to configured minimum and maximum bounds.

2.8 PPO Objective

$$(14) \quad \rho_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{old}}(a_t | s_t)$$

Here $\rho_t(\theta)$ is the PPO probability ratio, π_θ is the current policy, $\pi_{\theta_{old}}$ is the behavior policy used to collect the rollout, a_t is the action, and s_t is the state.

$$(15) \quad L_{CLIP}(\theta) = E_t[\min(\rho_t(\theta) A_{hat_t}, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) A_{hat_t})]$$

Here θ denotes policy parameters, $\rho_t(\theta)$ is the PPO probability ratio, A_{hat_t} is the advantage estimate, and ϵ is the clipping parameter. The report uses this only as a concise training-objective reference.

PPO is used because the thesis needs a stable, widely used policy-gradient algorithm for stochastic sequential control, not because it claims PPO is the best possible market-making RL algorithm. The action is discrete but two-dimensional MultiDiscrete: the agent chooses half-spread h and skew m , which together form a 25-action quote grid. DQN would require value-based handling of this grid and is less natural for the policy-gradient control framing used here; SAC is more naturally aligned with continuous actions and would expand the scope; and A2C is simpler but often less stable or sample-efficient than PPO. PPO therefore provides a reliable baseline algorithm for controlled ablation of the observation channels.

3 Use-Case Scenario

3.1 Synthetic HFMM Scenario

The use case is a synthetic market maker posting one bid and one ask quote at each step. The environment supplies mid-price dynamics, volatility estimates, and regime labels; the agent receives stochastic fills and is evaluated on wealth, risk-adjusted performance, inventory tails, and fill behavior.

3.2 Trading Agents and Strategy Variants

Table 3.1 Strategy variants

Strategy or variant	Observed information	Purpose
Naive	No learned state dependence	Fixed-spread baseline.
Avellaneda-Stoikov	Inventory, volatility, horizon	Analytical inventory-risk baseline.
ppo_aware	$\sigma_{\hat{}}$ and estimated regime one-hot	Original regime-aware PPO.
ppo_blind	$\sigma_{\hat{}}$ without regime one-hot	Original regime-blind PPO.
sigma_only	Continuous $\sigma_{\hat{}}$ only	Tests whether $\sigma_{\hat{}}$ alone carries the signal.
combined	$\sigma_{\hat{}}$ and estimated regime one-hot	Tests estimated categorical incremental value.
oracle_full	$\sigma_{\hat{}}$ and true regime one-hot	Tests perfect-label incremental value.
regime_only / oracle_pure	Categorical labels without $\sigma_{\hat{}}$	Anchor label-only variants.

3.3 Information-Design Question

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy.

4 System Architecture

4.1 Run Lifecycle and Reproducibility Layer

``run.py`` dispatches jobs from JSON configuration files, creates a timestamped run directory, snapshots the config, records git metadata, writes logs and metrics, and finalizes run status. Long WP6 jobs support resume validation against the saved config snapshot.

4.2 Synthetic Market and Detector Layer

``src/wp1/sim.py`` implements the fill and inventory simulator. ``src/wp2/synth_regime.py`` generates synthetic regime paths, mid-prices, rolling realized volatility, and detector outputs.

4.3 Gymnasium Environment Layer

``src/wp3/env.py`` wraps the simulator as a Gymnasium environment. The observation vector is $[q_{\text{norm}}, \sigma_{\hat{}}, \tau, \text{regime}_L, \text{regime}_M, \text{regime}_H]$. Warmup, invalid, or disabled regime labels produce a zero one-hot vector rather than an artificial medium label.

4.4 Strategy and Training Layer

The training layer fits PPO policies, the main evaluation layer compares strategies and ablations, and the signal-informativeness layer tests whether regime labels gain value as $\sigma_{\hat{}}$ is degraded. The fixed-spread and AS baselines provide non-learning comparators.

4.5 Evaluation and Evidence Layer

Evaluation writes CSV metrics, figures, and summaries. The thesis_34 adaptation embeds existing figures and quotes frozen numerical evidence; it does not regenerate protected evidence artifacts.

4.6 Methodology Pipeline Illustration

Figure 4.1 summarizes the evidence pipeline at a methodological level. A synthetic path generates the latent volatility regime and mid-price sequence; rolling realized volatility and detector labels form the observed signals; the Gymnasium environment converts these signals, inventory, time, quotes, and fills into sequential observations and rewards; PPO variants and analytical baselines are then evaluated on held-out out-of-sample metrics. The figure is included only to clarify the workflow and is not a regenerated performance artifact.



Methodology illustration only; no performance evidence or protected experiment output is regenerated.

Figure 4.1. Methodology-only HFMM evaluation pipeline. Synthetic market paths produce $\sigma_{\hat{}}$ and $\text{regime}_{\hat{}}$ signals, which feed the Gymnasium environment, PPO variants, baselines, and out-of-sample metrics. Generated by `scripts/figures/gen_pipeline_architecture_figure.py`; this illustration is not performance evidence.

5 Layered Experimental Model

Section 4 describes the software/evidence architecture, while this section restates the same workflow as a conceptual layered experimental model.

5.1 Layer 1: Market State Generation

Generate Markov regimes and mid-price paths under controlled volatility multipliers.

5.2 Layer 2: Signal Construction

Construct $\sigma_{\hat{}}$ from rolling realized volatility and derive estimated regime labels.

5.3 Layer 3: Observation Encoding

Encode inventory, volatility, time-to-horizon, and optional regime one-hot channels.

5.4 Layer 4: Action and Execution

Decode PPO actions into bid/ask quote distances and sample Poisson-arrival fills.

5.5 Layer 5: Learning and Evaluation

Train PPO on the chronological train segment and evaluate deterministic policies OOS.

6 Algorithm Specification and Pseudocode

6.1 Plain-Language Overview

The experiment creates a synthetic market path, estimates volatility signals, trains policies on the first 70% of the path, and evaluates all strategies on the held-out 30%. Ablations modify which volatility channels the PPO policy can observe.

6.2 Synthetic Market and Regime Generation

Algorithm 1: Synthetic market and regime generation

```

Input: seed, transition matrix P, base volatility, volatility multipliers, n_steps
Initialize  $z_0$  and  $S_0$ 
For  $t = 0 \dots n\_steps - 1$ :
    sample  $z_{t+1}$  from  $P[z_t]$ 
    set  $\sigma_t = \text{base\_sigma} * \text{multiplier}[z_t]$ 
    sample mid-price increment and update  $S_{t+1}$ 
Compute rolling  $\sigma_{hat}$  after warmup window
Calibrate thresholds on warmup data and assign  $\text{regime}_{hat}$ 
Output: exogenous table with mid,  $\sigma_{hat}$ ,  $\text{regime}_{true}$ ,  $\text{regime}_{hat}$ 

```

6.3 Market-Making Environment Step

Algorithm 2: Market-making environment step

```

Input: action ( $h_{idx}$ ,  $m_{idx}$ ), current simulator state, exogenous row
Decode  $h = h_{idx} + 1$  and  $m = m_{idx} - 2$ 
Set  $\Delta_{bid} = \max(1, h + m)$ ,  $\Delta_{ask} = \max(1, h - m)$ 
Compute fill intensities and per-step fill probabilities
Sample bid and ask fills
Update inventory and cash, including fees
Advance mid-price and compute  $W_{t+1}$ 
Return observation, reward =  $\Delta_{equity} - \eta * \text{inventory}^2$ , done flag, info

```

6.4 PPO Training and OOS Evaluation Protocol

Algorithm 3: PPO training and OOS evaluation protocol

```

For each seed and variant:
    generate one exogenous synthetic path

```

```

split path chronologically into train and test segments
configure observation channels for the variant
train PPO on the train segment for the configured timesteps
evaluate the deterministic policy on the test segment
log Sharpe-like ratio, final equity, inventory p99, fill rate, and per-regime metrics
Aggregate seed-paired statistics across variants

```

6.5 Experiment 5: Signal-Informativeness Sweep

Algorithm 4: Signal information-value sweep

```

For each degradation condition in {full, noisy, lagged, coarsened, none}:
    transform sigma_hat according to the condition
    for each valid variant and seed:
        train PPO and evaluate OOS
Aggregate condition-variant means and paired comparisons
Test whether combined gains value as sigma_hat is degraded

```

6.6 Reference Implementation Map

Table 6.1 Reference implementation map

Component	Reference file
Run lifecycle	src/run_context.py, run.py
Simulator	src/wp1/sim.py
Regime generation and detectors	src/wp2/synth_regime.py
Gymnasium environment	src/wp3/env.py
PPO training	src/wp4/job_w4_ppo.py
WP5 evaluation	src/wp5/job_w5_eval.py
WP6 sweep	src/wp6/job_w6_sweep_full.py

6.7 Complexity Analysis

Simulation and deterministic evaluation are linear in the number of environment steps. PPO training cost is approximately linear in total timesteps, policy-network forward/backward passes, and the number of seed-variant cells. The WP6 full sweep is expensive because it multiplies conditions, variants, and seeds; this is why its outputs are treated as frozen evidence.

The implementation prioritizes reproducibility and auditability over maximum simulation throughput. Future versions could improve runtime through vectorized environments, parallel rollouts, or lower-level simulator components for the execution loop.

6.8 Executable Example Commands

Commands

```
python run.py --config config/w5_main.json
python run.py --config config/w5_detector_full.json
python run.py --config config/w5_eta_regime.json
python run.py --config config/w5_misspec_mild.json
python run.py --config config/w6_sweep_full.json
python run.py --config config/w6_sweep_full.json --resume <run_id>
```

7 Experimental Setup

7.1 Canonical Configuration Protocol

Canonical experiments are driven by JSON configs under `config/`. Shared parameters include $\text{mid0} = 100.0$, $\text{tick_size} = 0.01$, $\text{dt} = 0.2$, $\text{baseline sigma_mid_ticks} = 0.8$, $A = 5.0$, $k = 1.5$, $\text{fee_bps} = 0.2$, and $\text{latency_steps} = 1$.

Market context. The simulator is asset-class agnostic and uses normalized price units. It should be read as a stylized liquid electronic limit-order-book market rather than as a calibrated model of a specific venue such as equities, futures, crypto, or BIST. The fee and latency parameters are controlled friction assumptions used to make strategy comparisons internally consistent. They are not intended to identify a particular exchange microstructure.

7.2 Data Generation and Preprocessing

Each run generates synthetic mid-price and regime paths, computes rolling realized volatility, and assigns detector labels. The main reported pipelines use causal rv_baseline labels.

7.3 Train/Test Split

The main out-of-sample evaluation and signal-informativeness sweep use a chronological 70/30 split on the exogenous series. PPO trains on the first segment and is evaluated deterministically on the held-out OOS segment.

7.4 Strategy Variants

Experiment 1, the Main Out-of-Sample Evaluation, evaluates naive, AS, ppo_aware , and ppo_blind . Experiment 2, the Five-Variant Signal Ablation, and Experiment 5, the Signal-Informativeness Sweep, use sigma_only , regime_only , combined , oracle_pure , and oracle_full .

7.5 Detector Variants

Table 7.1 Detector variants

Detector	Role	Caveat
rv_baseline	Main causal detector	Rolling RV threshold; used in main WP4/WP5/WP6 pipelines.
rv_dwell	Auxiliary robustness detector	Offline dwell smoothing; not the main causal detector.
hmm	Robustness detector	Higher classification accuracy but no reliable PPO advantage.

7.6 PPO Hyperparameters

7.6.1 Hyperparameter Selection Protocol

The thesis does not perform a broad hyperparameter optimization campaign. This is intentional because the central comparison is signal-channel design, not PPO tuning.

PPO hyperparameters are held fixed across PPO variants to avoid giving one information condition a tuning advantage.

$\eta = 0.001$ is the canonical inventory-penalty scale selected after the earlier inventory-penalty ablation.

No separate validation set is introduced because the study uses a fixed train/OOS evaluation protocol; this should be treated as a limitation and a future extension if full tuning is pursued.

This is a methodological choice, not a claim that the PPO settings are globally optimal.

Table 7.2 PPO hyperparameters

Hyperparameter	Canonical full-run value
total_timesteps	1,000,000
learning_rate	$3e-4$
n_steps	2048
batch_size	256
n_epochs	10
gamma	0.999
clip_range	0.2
ent_coef	0.01 for full WP5/WP6 runs
η	0.001

7.7 Protected Evidence and No-Rerun Policy

The thesis_35 adaptation reuses frozen performance results. It does not rerun PPO training, WP5/WP6 experiments, detector robustness, ablations, misspecification checks, protected CSV generation, or frozen performance-figure generation. The only newly generated visuals are methodology-only illustrations produced by ``scripts/figures/gen_synthetic_environment_figure.py`` and ``scripts/figures/gen_pipeline_architecture_figure.py``.

8 Results and Visualisation

8.1 Experiment 1: Main Out-of-Sample Evaluation

Experiment 1, the 20-seed Main Out-of-Sample Evaluation, shows that PPO variants produce much higher risk-adjusted performance than the fixed-spread and AS baselines. AS can produce higher raw equity, but with substantially larger inventory exposure.

Table 8.1 Main OOS strategy metrics

Strategy	Sharpe-like	Final equity	inv_p99	Fill rate
ppo_aware	0.715	4.10	2.00	0.236
ppo_blind	0.740	4.42	2.05	0.232
AS	0.105	5.05	29.95	0.444
naive	0.127	4.49	21.20	0.119

AS has higher raw equity but carries substantially larger inventory tail risk; therefore the main comparison is risk-adjusted performance and inventory control, not raw equity alone.

The AS implementation is used as a canonical analytical inventory-aware comparator rather than as an exhaustively optimized trading system. The PPO-versus-AS comparison establishes a reference point for learned quoting and inventory control, but the central thesis claim is the within-PPO signal-design comparison between $\sigma_{\hat{}}$, estimated labels, and oracle labels.

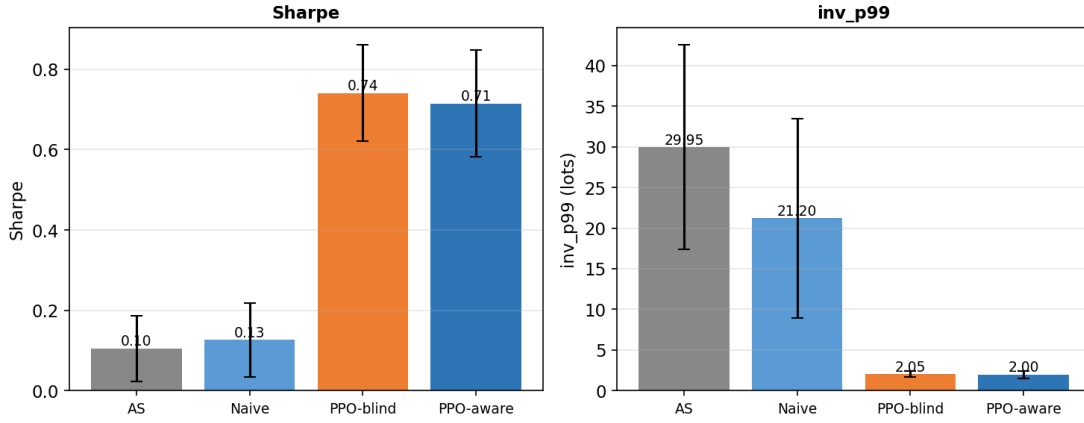


Figure 8.1. Experiment 1 main OOS results: PPO variants dominate naive and AS on risk-adjusted performance, while AS carries much larger inventory tail risk.

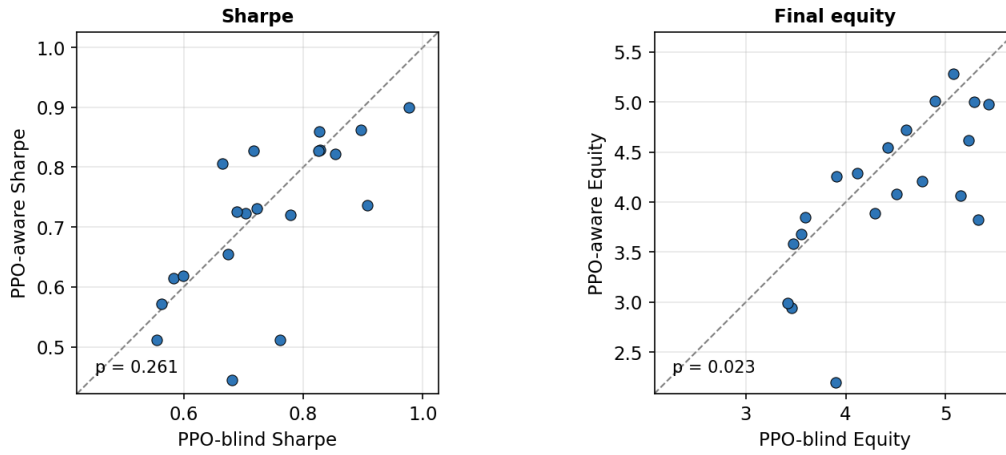


Figure 8.2. Seed-paired PPO-aware versus PPO-blind comparison: the estimated regime channel does not create a robust Sharpe-like advantage.

The paired-seed view shows that the aware policy does not dominate the blind policy seed by seed. The equity panel even favors the blind policy in the canonical paired test.

8.2 Five-Variant Ablation

Table 8.2 Five-variant ablation metrics

Variant	Signals	Sharpe-like	Final equity	inv_p99
ppo_sigma_only	sigma_hat only	0.753	4.548	1.950
ppo_oracle_full	sigma_hat + regime_true	0.722	4.068	2.050
ppo_regime_only	regime_hat only	0.698	4.005	2.150

ppo_combined	sigma_hat + regime_hat	0.696	3.905	1.800
ppo_oracle_pure	regime_true only	0.684	3.925	1.950

The strongest ablation result is that `sigma_only` has the highest mean Sharpe-like value, while `oracle_full` does not significantly beat it. TOST supports practical equivalence under the ± 0.10 Sharpe-like bound.

The label-quality objection is also bounded by the `oracle_pure` result. `oracle_pure` receives the true categorical regime label but not `sigma_hat`, yet it remains below `sigma_only` in mean Sharpe-like performance in the five-variant ablation. This suggests that the limitation is not only detector noise: in this calibration, the continuous volatility signal is more useful for quote control than the discretized regime category.

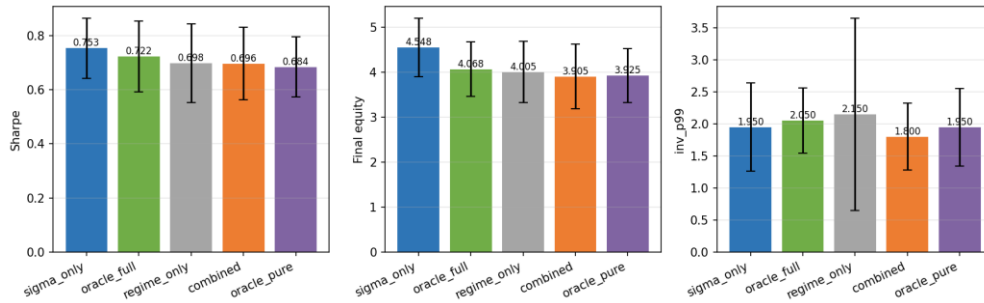


Figure 8.3. Five-variant ablation summary: `sigma_only` is the strongest mean Sharpe-like variant, and adding categorical labels does not improve it.

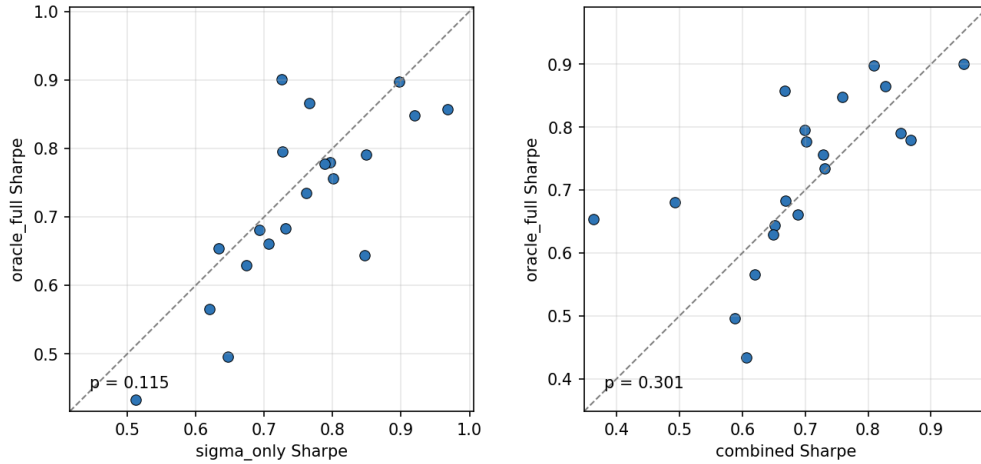


Figure 8.4. Oracle-label paired-seed comparison: even true regime labels do not reliably improve on `sigma_hat` alone.

The compact statistical summary below collects the canonical paired tests used to interpret the ablation and robustness results.

Table 8.3 Compact statistical summary

Comparison	Metric or test	Canonical result	Interpretation
PPO-aware vs PPO-blind	Sharpe-like paired t-test	$p = 0.261$	No significant Sharpe improvement.
PPO-aware vs PPO-blind	Final equity paired t-test	$p = 0.023$	Favors PPO-blind.
ppo_sigma_only vs ppo_oracle_full	Sharpe-like paired t-test	$p = 0.115$	No oracle-label Sharpe improvement.
ppo_sigma_only vs ppo_oracle_full	TOST +/-0.10	$p = 0.00067$; 90% CI [-0.001, +0.063]	Practical equivalence supported.
HMM detector	Detector accuracy	81.8%	Higher accuracy does not create advantage.
Detector robustness	ANOVA	$p = 0.997$	Detector choice does not explain result.
Regime-conditional eta	sigma_only vs combined Sharpe	$p = 0.0016$	Favors sigma_only.
Mild misspecification	sigma_only vs oracle_full Sharpe t-test	$p = 0.881$	No significant Sharpe difference.
Mild misspecification	TOST +/-0.05	$p = 0.042$	Practical equivalence supported.

The t-test results show no significant Sharpe improvement from explicit regime labels, while the TOST result provides positive evidence of practical equivalence under the stated bound.

8.3 Detector Robustness

Table 8.4 Detector robustness summary

Detector	Aware vs blind conclusion	Key statistic
rv_baseline	No reliable aware Sharpe advantage	$p = 0.1142$
rv_dwell	No reliable aware Sharpe advantage	$p = 0.1095$
hmm	No reliable aware Sharpe advantage despite higher accuracy	$p = 0.0822$
ANOVA	Detector choice does not explain the result	$F = 0.0034$, $p = 0.9966$

The ANOVA statistic is retained as a descriptive robustness summary across detector-specific PPO-aware Sharpe values. Because the same seeds are shared across detector conditions, the primary inferential evidence remains the seed-paired detector-specific tests; the ANOVA should not be read as the sole formal test of a repeated-measures design.

The detector robustness table comes from Experiment 3, the Detector Robustness experiment (w5_detector_full), while the main OOS table in Section 8.1 comes from Experiment 1 (w5_main). Small differences in PPO-aware/blind means therefore reflect distinct frozen experiment batches, not an inconsistency.

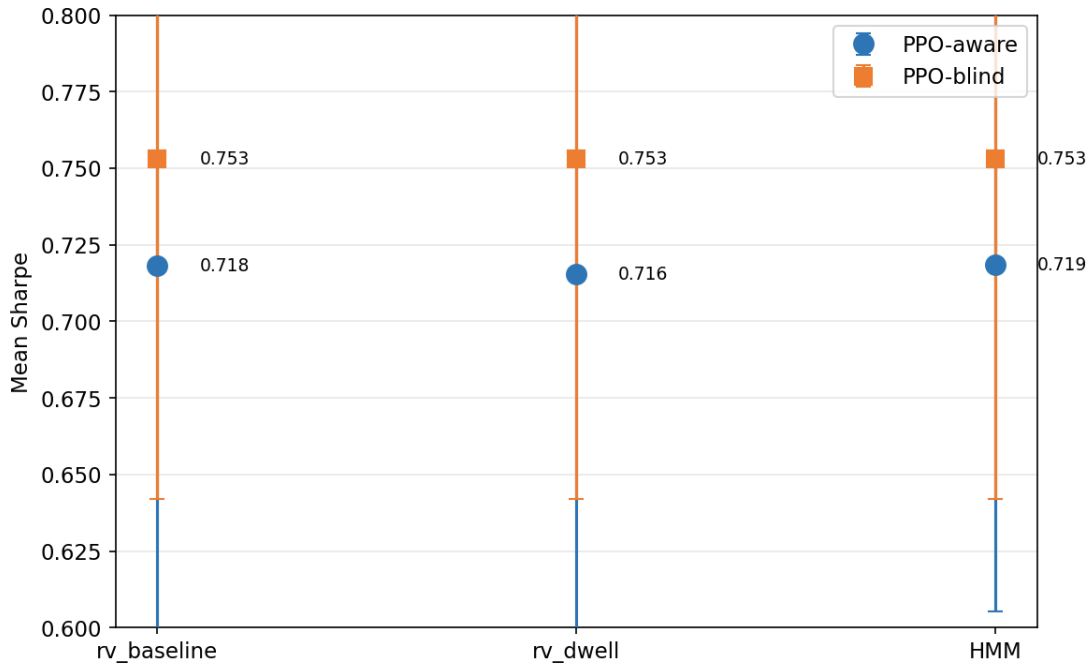


Figure 8.5. Detector robustness: *rv_baseline*, *rv_dwell*, and *HMM* all fail to create a reliable regime-aware PPO advantage.

8.4 Reward-Shaping and Misspecification Checks

The regime-conditional eta run tests whether explicit labels become useful when the reward penalizes high-volatility inventory more strongly. *sigma_only* still beats combined on Sharpe-like performance ($p = 0.0016$). Under mild regime-dependent execution misspecification, *sigma_only* and *oracle_full* remain statistically indistinguishable and practically equivalent under the reported TOST bound.

In the misspecification check, the mean difference between *sigma_only* and *oracle_full* is very small, but the ± 0.05 TOST result is close to the threshold; it should therefore be read as supportive but less decisive than the main five-variant ablation equivalence result.

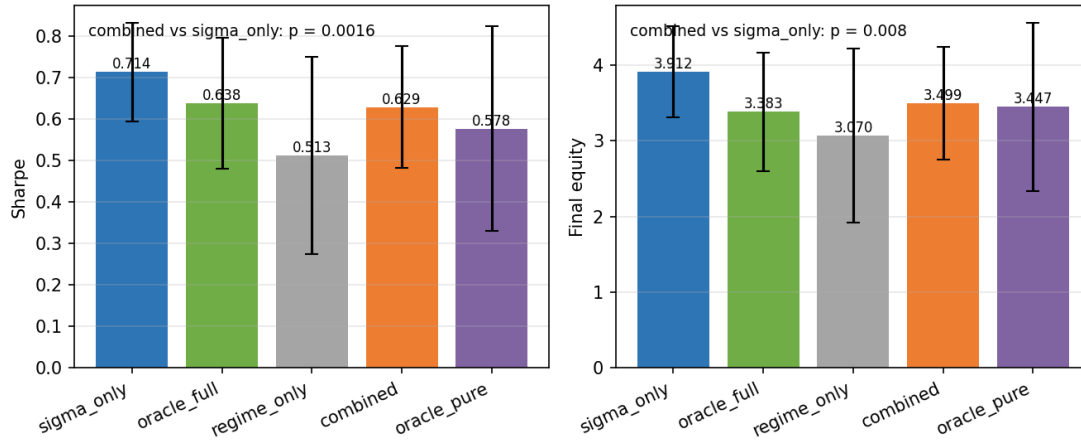


Figure 8.6. Regime-conditional eta summary: changing the reward channel by regime still favors sigma_only over combined.

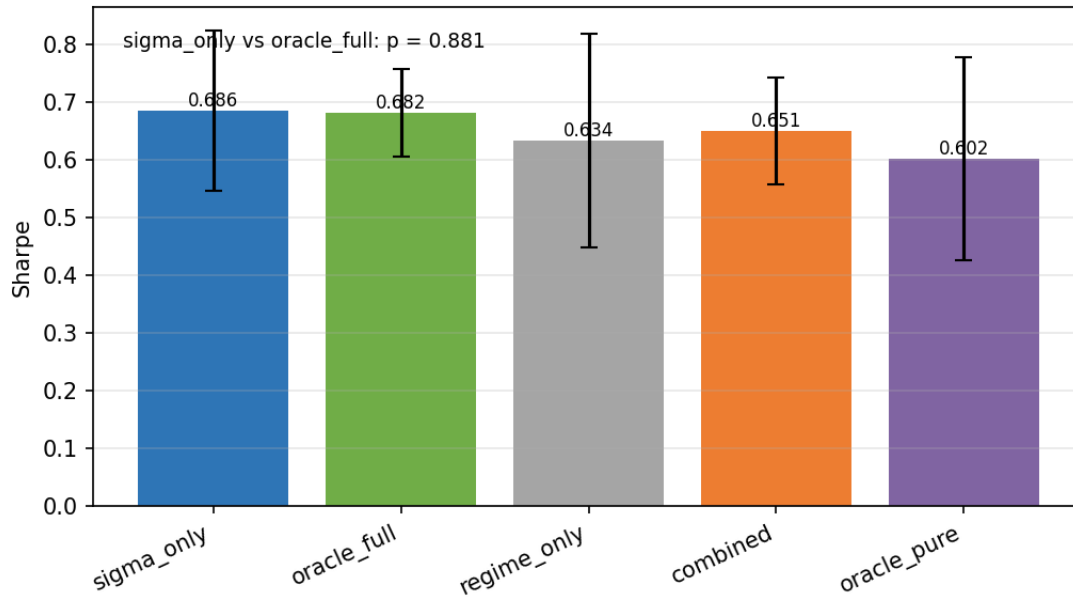


Figure 8.7. Mild model-misspecification summary: sigma_only and oracle_full remain close under regime-dependent execution parameters.

8.5 Experiment 5: Signal-Informativeness Sweep

Table 8.5 Signal information-value sweep

condition	sigma_only	combined	regime_only	oracle_full	oracle_pure
full	0.7626 +/- 0.0596	0.6898 +/- 0.0607	0.6991 +/- 0.0668	0.6811 +/- 0.0697	0.6632 +/- 0.0769
noisy	0.7563 +/- 0.0616	0.7125 +/- 0.0628	0.6991 +/- 0.0668	0.6736 +/- 0.1013	0.6632 +/- 0.0769
lagged	0.7822 +/- 0.0454	0.6812 +/- 0.0536	0.6991 +/- 0.0668	0.7264 +/- 0.0686	0.6632 +/-

					0.0769
coarsened	0.7827 +/- 0.0533	0.6908 +/- 0.0827	0.6991 +/- 0.0668	0.7447 +/- 0.0549	0.6632 +/- 0.0769
none	undefined	0.6991 +/- 0.0668	0.6991 +/- 0.0668	0.6632 +/- 0.0769	0.6632 +/- 0.0769

Entries are mean +/- 95% confidence-interval half-width across 20 seeds, not standard deviations.

Experiment 5, the Signal-Informativeness Sweep, did not support the original informativeness-threshold hypothesis within the tested calibration band. `sigma_only` remained high under informative conditions, while combined was directionally below `sigma_only`.

The Signal-Informativeness Sweep figures should be read as a refinement of the main finding rather than a new mechanism proof: they show that the tested `sigma_hat` degradation path did not reveal a regime-label advantage.

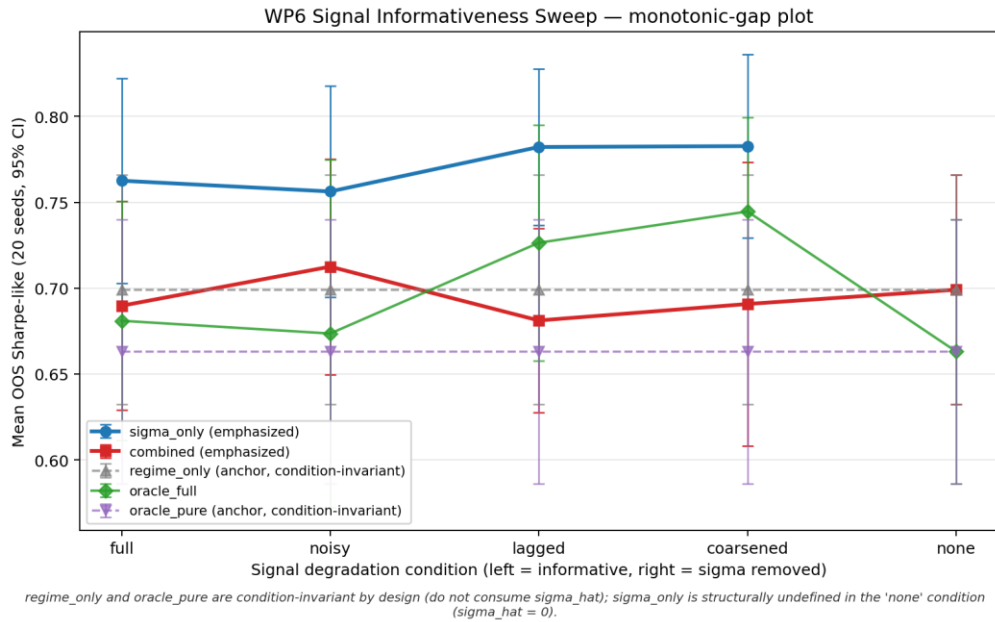


Figure 8.8. Experiment 5 monotonic-gap plot: the expected narrowing of the `sigma_only` versus combined gap does not appear in the tested calibration band.

One possible interpretation is that the degraded continuous signal still preserves enough ordinal volatility information for quote-width adaptation, whereas the categorical label is coarser and does not add useful variation in this calibration. In noisy or lagged settings, adding the label may also increase the observation dimension without improving the control-relevant signal. This should be read as a performance pattern in the tested calibration band, not as proof of the PPO policy's internal representation mechanism.

The strongest degradation pattern appears in the lagged condition, where combined is materially below `sigma_only` (mean difference about -0.101; Cohen's *d*z about -0.91). This suggests that a stale

categorical channel can be harmful when it adds delayed/coarse state information rather than control-relevant volatility information.

The paired-seed views below show whether aggregate differences are broad across seeds or driven by a few runs.

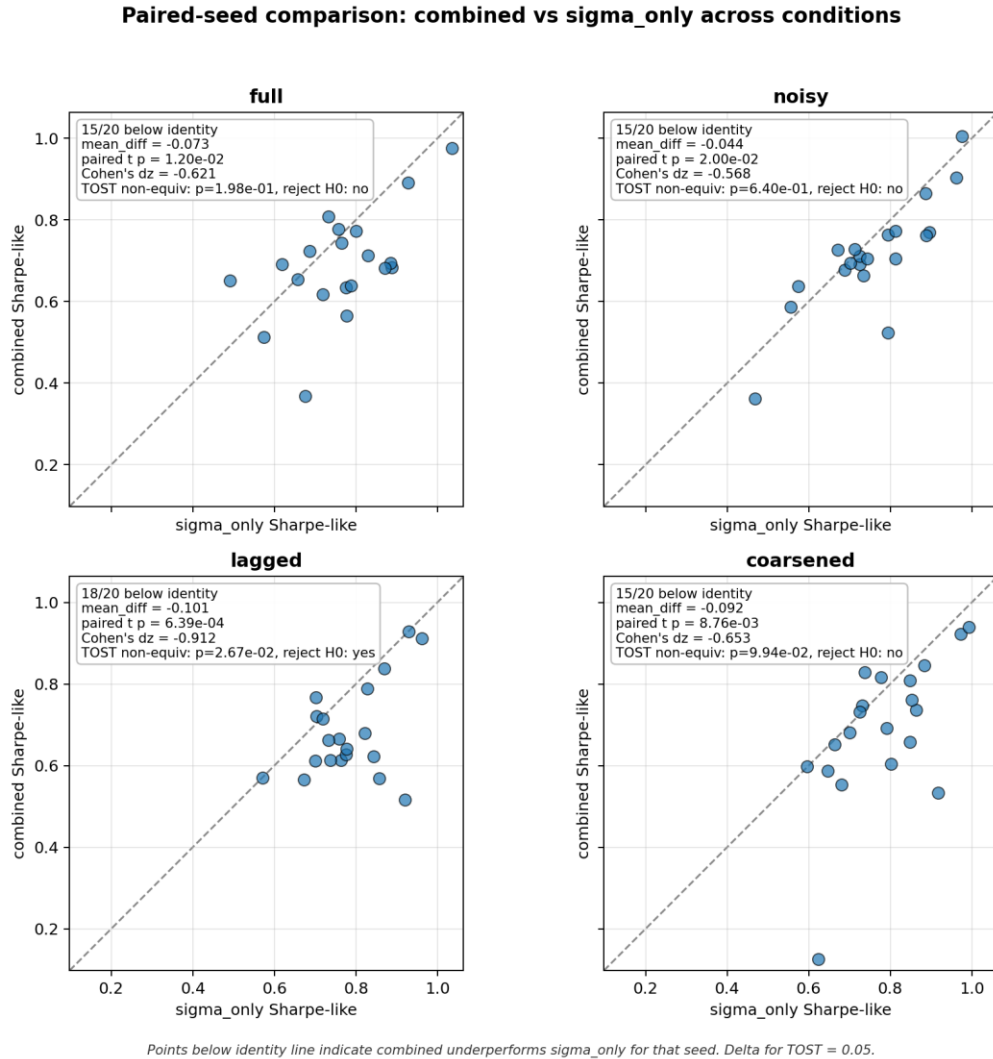


Figure 8.9. Experiment 5 paired-seed combined versus sigma_only: combined is directionally below sigma_only in informative conditions.

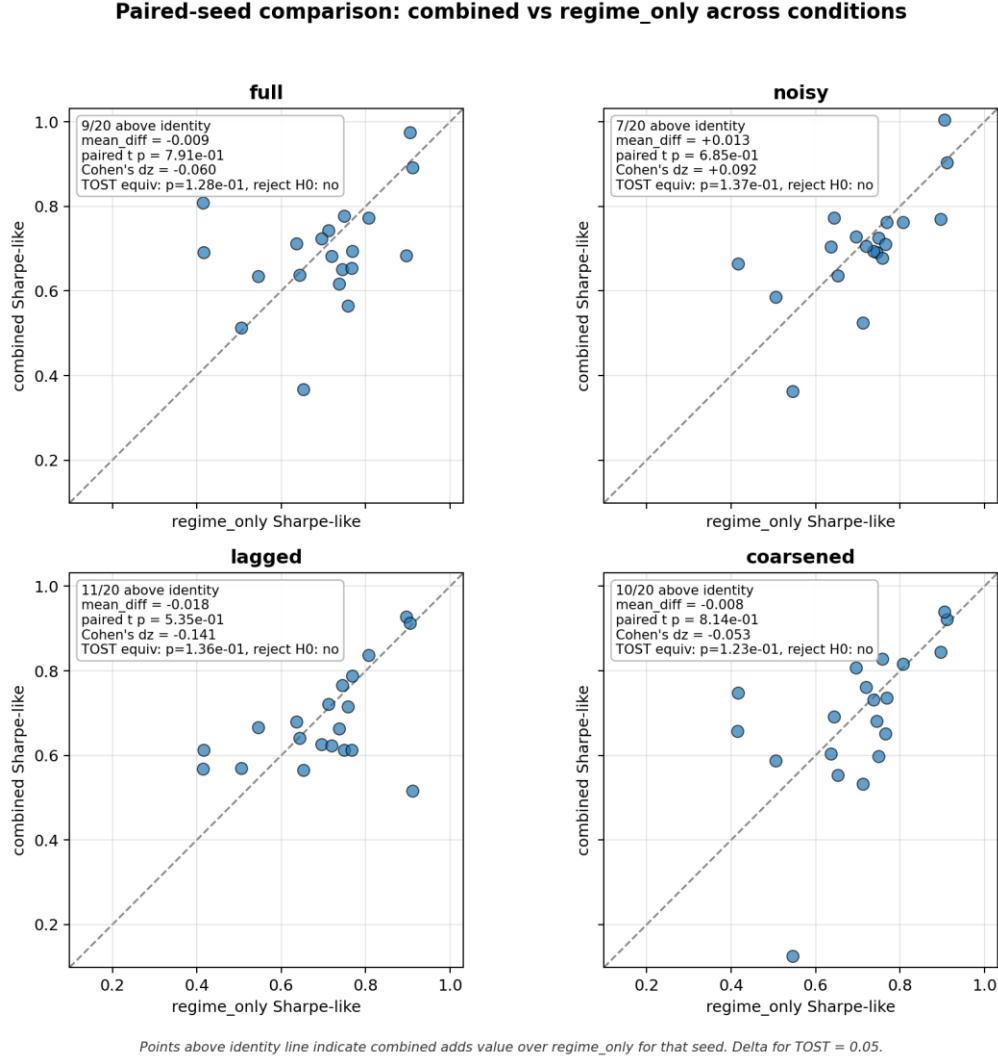


Figure 8.10. Experiment 5 paired-seed combined versus regime_only: this diagnostic bounds how much $\sigma_{\hat{}}$ is used inside the combined variant.

9 Evaluation Metrics

9.1 Final Equity

Final equity is the terminal marked-to-market wealth. It is economically meaningful but not sufficient alone because high raw equity can be earned by taking large inventory risk.

9.2 Sharpe-Like Ratio

$$(16) \text{ Sharpe-like} = \text{mean}(\Delta W_t) / \text{std}(\Delta W_t) * \sqrt{1 / dt}$$

Let $\Delta W_t = W_t - W_{t-1}$. The reported Sharpe-like metric is computed as $\text{mean}(\Delta W_t) / \text{std}(\Delta W_t)$, using the sample standard deviation with $\text{ddof} = 1$, multiplied by $\sqrt{1 / dt}$ when the standard deviation is positive; otherwise it is reported as 0. It is therefore a simulator-scale risk-adjusted PnL metric rather than an annualized market Sharpe ratio.

9.3 Inventory Tail Risk

$$(17) \text{ inv_p99} = \text{percentile_99}(|q_t|)$$

Here q_t is inventory. inv_p99 is a practical tail-risk diagnostic for whether a strategy earns returns by carrying large inventory.

9.4 Fill Rate

$$(18) \text{ fill_rate} = \text{filled quote opportunities} / \text{total quote opportunities}$$

The numerator and denominator follow the logged project convention. Fill rate is diagnostic rather than the primary thesis metric.

9.5 Paired t-Tests

$$(19) t = \text{mean}(d_i) / (\text{sd}(d_i) / \text{sqrt}(n))$$

Here d_i is the paired seed-level difference between two strategies, $\text{sd}(d_i)$ is its sample standard deviation, and n is the number of paired seeds.

Because the manuscript reports several paired comparisons, isolated marginal p-values should be read descriptively rather than as a stand-alone familywise discovery claim. The main conclusion is based on the repeated pattern across the main OOS evaluation, detector robustness, oracle ablations, reward-shaping, misspecification, and signal-informativeness diagnostics.

9.6 TOST Equivalence Tests

TOST evaluates whether the plausible range of a paired difference lies inside a pre-specified practical-equivalence band. At $\alpha = 0.05$, the corresponding equivalence reading uses the 90% confidence interval. The Five-Variant Signal Ablation uses TOST to support practical equivalence for sigma_only versus oracle_full ; the Signal-Informativeness Sweep also uses related paired tests to examine non-equivalence and mean indistinguishability patterns.

The equivalence bounds should be read as practical smallest-effect-size thresholds for the Sharpe-like metric, not as universal constants. A wider ± 0.10 band is used for the main five-variant ablation, where the question is whether oracle regime information creates a practically meaningful improvement over sigma_hat alone. Tighter ± 0.05 bounds are used in narrower robustness or misspecification checks. The thesis therefore does not treat TOST as a stand-alone proof; it reports TOST together with paired tests, confidence intervals, and directional seed-level evidence.

10 Discussion

10.1 Main Interpretation: Signal Redundancy

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once sigma_hat is already observed by the PPO policy. The results are consistent with signal redundancy.

The interpretation is that the continuous realized-volatility proxy already supplies the economically useful volatility information needed by the policy in this environment. The categorical label may be coarser or redundant when presented alongside $\sigma_{\hat{}}$.

The evidence should therefore be read slightly more precisely than pure redundancy. If a categorical label only repeated the useful information in $\sigma_{\hat{}}$, combined and σ_{only} would be expected to be approximately indistinguishable. In several experiments, however, combined is directionally below σ_{only} . This pattern is consistent with signal redundancy plus mild categorical-channel degradation: the label may be coarser than the continuous volatility estimate, may add observation dimensionality, or may interact with policy learning without adding control-relevant information. This is an interpretation of the observed performance pattern, not a proof of the PPO policy's internal mechanism.

10.2 Why the Result Is Not a Weak Null

The conclusion does not rest only on $p > 0.05$. It is supported by detector robustness, oracle labels, TOST equivalence, reward-channel checks, mild misspecification, and signal-degradation tests.

10.3 Synthetic-Market Boundary

The result is bounded to the tested synthetic HFMM simulator. The simulator deliberately isolates quote distance, Poisson-arrival fills, fees, latency, inventory, and volatility regimes so that the incremental information value of the regime channel can be tested cleanly.

Real limit-order books include additional mechanisms that are outside the main simulator: queue position, execution priority, adverse selection, order-book imbalance, richer latency effects, clustered or self-exciting order arrivals, non-stationary liquidity, and exchange-specific transaction-cost and microstructure rules. These mechanisms can change both the value of volatility information and the way a categorical regime label interacts with execution risk.

For that reason, the thesis should be read as controlled synthetic evidence, not as a real-market external validity claim. Richer LOB simulators, Hawkes-process LOB models, queue-reactive execution, and real-data evaluation are natural future work and could change the incremental value of regime labels.

The out-of-sample protocol should also be interpreted carefully. It is a chronological 70/30 temporal hold-out within the same synthetic data-generating process, not a distributional-shift test across fundamentally different markets. Multiple seeds improve robustness to simulation randomness, but they do not establish external validity beyond the controlled synthetic environment.

10.4 Risks to Interpretation

Table 10.1 Risks to interpretation

Risk	Mitigation
Overclaiming mechanism	Use signal-redundancy and categorical-channel degradation as interpretations, not proofs of PPO internals.
Detector causality confusion	State that rv_{baseline} is the main causal detector and rv_{dwell} is auxiliary/offline.

Synthetic external validity	Frame all claims as controlled synthetic-market claims.
Template over-compression	Keep the literature and evidence spine, but map it into template headings.
Mid-price mark-to-market assumption	Report it as a simulator evaluation convention; real liquidation would require spread/depth/queue/adverse-selection considerations.

10.5 Future Work

Evaluate stronger regime-dependent execution misspecification and richer non-stationary liquidity settings.

Test richer limit-order-book simulators with queue position, order-book imbalance, latency priority, adverse selection, and self-exciting order flow.

Use Hawkes-process LOB models and real-data evaluation after thesis scope, not as current claims of this report.

Run representation-level diagnostics only as future mechanism work.

Explore alternative observation encodings and policy architectures.

11 Reproducibility Checklist

The reproducibility path uses a local Python virtual environment created with `python -m venv .venv` and dependencies installed from `requirements.txt`.

Main commands are documented in Table 11.3. Exact reproducibility depends on using the project repository, config snapshots, fixed seeds, logged run metadata, and the protected evidence files recorded in `EVIDENCE_MANIFEST.md`.

Protected evidence is checked by SHA256. No Dockerfile or `environment.yml` is present, so this report does not claim Docker or Conda reproducibility.

No external CI service is claimed; validation is documented through local reproducibility checks and protected evidence hashes.

Formal line-coverage percentage is not claimed; the tests are targeted regression and reproducibility checks.

Table 11.1 Reproducibility checklist

Item	Status in thesis_35
Frozen baseline	thesis-v29-frozen tag; commit 9681faa; thesis_29 remains untouched.
Template draft	thesis_35 is a final format-guide compliance polish draft built on thesis_34; frozen evidence remains unchanged.

Run ID structure	YYYYMMDD-HHMMSS_seed<S>_<TAG>_<COMMIT>.
Config snapshots	Every run snapshots config; config_snapshot_all.md inventories active config files.
Seeds	WP5 main uses seeds 1-20; WP6 full uses seeds 42-61.
Train/test split	70/30 chronological split on exogenous synthetic series.
Evidence manifest	EVIDENCE_MANIFEST.md records protected evidence and remediation invariants.
Protected CSV hashes	Listed below and verified by file hash checks.
Codebase snapshot	docs/internal/codebase_snapshot.py.
No-rerun policy	No PPO/WP5/WP6 reruns and no frozen evidence-figure regeneration; only the methodology-only synthetic-environment illustration was generated.

Table 11.2 Protected artifact hash checks

Protected artifact	Expected SHA256	Check
results/metrics_detector_compare.csv	28E7AD40BB47...	MATCH
docs/internal/wp6_sweep_full/summary_condition_variant.csv	6DD627E81637...	MATCH
docs/internal/wp6_sweep_full/summary_paired_combined_vs_sigma.csv	4BABCAAACE1D...	MATCH
docs/internal/wp6_sweep_full/summary_paired_combined_vs_regime.csv	2087FEFBE5DC...	MATCH

Table 11.3 Reproducibility commands

Reproducibility command	Purpose
python -m venv .venv	Create environment.
& .\venv\Scripts\Activate.ps1	Activate environment on Windows PowerShell.
pip install -r requirements.txt	Install dependencies.
python run.py --config config/w5_main.json	Run main WP5 evaluation.
python run.py --config config/w6_sweep_full.json	Run WP6 sweep.
python run.py --config config/w6_sweep_full.json --	Resume WP6 sweep.

resume <run_id>	
ruff check src/	Lint source code.

These commands document reproducibility paths only; thesis_35 adaptation did not rerun experiments.

12 Conclusion

This template-adapted report preserves the thesis finding while reorganizing the manuscript into the requested technical-report structure. PPO learns strong risk-adjusted quoting behavior in the synthetic HFMM simulator, but the explicit categorical regime channel does not provide robust incremental value beyond $\sigma_{\hat{\theta}}$. The result is best defended as evidence consistent with signal redundancy in the tested controlled environment.

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<https://doi.org/10.1177/2515245918770963>

Appendix A Code Appendix

Appendix Table A.1 Code map

Area	Files
Run management	run.py; src/run_context.py
Simulation	src/wp1/sim.py; src/wp1/w1_as_baseline.py
Regimes	src/wp2/synth_regime.py; src/wp2/job_w2_synth.py
Environment	src/wp3/env.py; src/wp3/w3_sanity.py
PPO training	src/wp4/job_w4_ppo.py
Evaluation	src/wp5/job_w5_eval.py; src/wp5/job_w5_detector_compare.py
Signal audit and sweep	src/wp5_5/*; src/wp6/job_w6_sweep_full.py

Canonical commands

```
python run.py --config config/w5_main.json
python run.py --config config/w5_detector_full.json
python run.py --config config/w5_eta_regime.json
python run.py --config config/w5_misspec_mild.json
python run.py --config config/w6_sweep_full.json
```

Appendix B Unit Tests and Validation Checks

Appendix Table B.1 Unit test and validation checklist

Check	Purpose
-------	---------

tests/test_csv_metric_logger.py	Verifies stable CSV metric schema behavior.
tests/test_resume_validation.py	Verifies resume-mode config validation.
WP3 sanity checks	Exercises naive, AS, and random policies through the Gymnasium environment.
Figure/path checks	Ensures thesis_35 references existing frozen figure files.
Forbidden phrase scan	Guards against overclaims in the generated draft.

No external CI service is claimed; validation is documented through local reproducibility checks and protected evidence hashes.

Formal line-coverage percentage is not claimed; the tests are targeted regression and reproducibility checks.

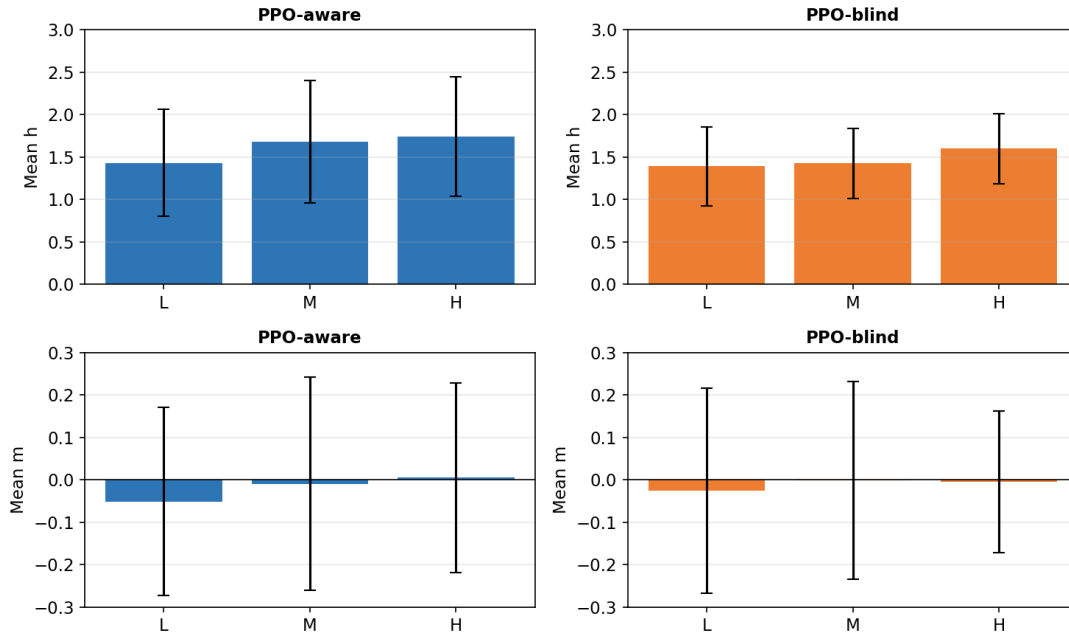
Pass/fail criteria: all pytest tests pass; generated artifacts exist and are non-empty; protected CSV hashes match EVIDENCE_MANIFEST.md; no forbidden overclaim phrases are detected; no experiment reruns are required for document generation.

Test categories include CSV metric logger schema stability, resume-mode config validation, WP3 environment sanity checks, figure/path existence checks, forbidden phrase or overclaim scans, and protected evidence SHA256 checks.

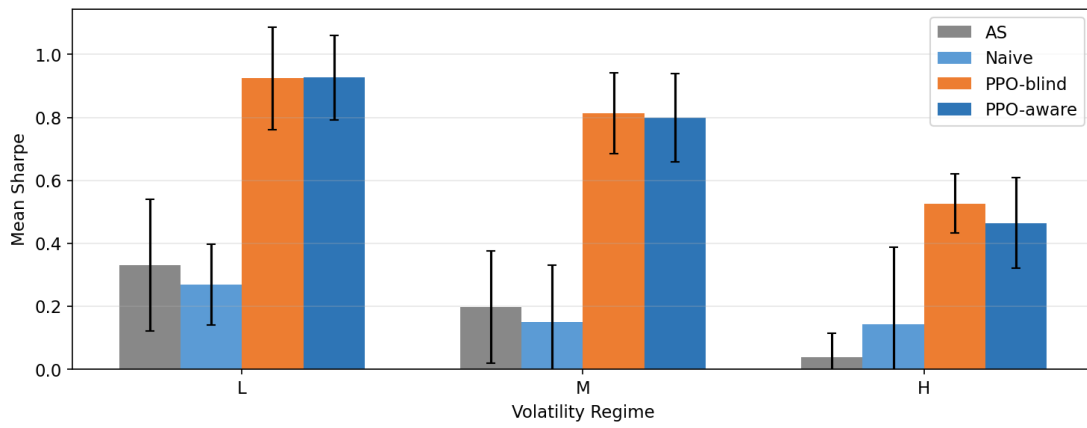
How to run tests: `pytest -q`; `pytest tests/test_csv_metric_logger.py -q`; `pytest tests/test_resume_validation.py -q`; `python run.py --config config/w3_sanity.json`.

Appendix C Extended Evidence Tables

This appendix contains diagnostic material demoted from the main body to keep the report template focused.



Appendix Figure C1. Regime-wise action distribution diagnostic.



Appendix Figure C2. Regime-wise Sharpe diagnostic.

The table below summarizes supporting diagnostics. These items support interpretation and provenance; they do not replace the canonical frozen evidence chain used for the main claims.

Appendix Table C.1 Supporting diagnostics

Diagnostic	Status	Interpretation
Post-hoc signal diagnostics	Supporting only	Consistent with signal redundancy but not primary evidence.
WP5.5 signal audit	Offline calibration	Supports WP6 degradation design; no PPO training.
Detector robustness statistics	Protected evidence	Addresses poor-detector objection without changing main detector caveat.
WP6 paired summaries	Protected evidence	Supports refined interpretation of the signal-informativeness sweep.

This separation keeps the appendix readable while preserving the boundary between primary experiment evidence and supporting diagnostics.

Appendix D Author Contribution Statement

This thesis is submitted as a single-author MSc thesis by Onur Emiroglu. The reported implementation, experiments, documentation, and interpretation are organized in the project repository and were prepared under normal thesis supervision. Supervisory feedback is reflected in the final framing and limitations, but no multi-author contribution matrix is required.